Microprocessor/Microcontroller Lab

DYNAMIC Industries

DIVMC-8509U DIVMC-8509ADU

8085 MICROPROCESSOR TRAINER WITH LCD DISPLAY, USB INTERFACE, ADC, DAC, EPROM PROGRAMMER & INBUILT POWER SUPPLY

TECHNICAL SPECIFICATIONS FOR VMC-8509U

- Based on 8085 CPU operating at 6.144 MHz crystal
- 16K bytes of EPROM with powerful monitor program
- 16K byte of RAM available to the user
- Battery backup for user's RAM with 3.6V rechargeable cell
- Total on board memory expansion upto 64K bytes using 2732/2764/27128/6264
- Memory mapping definable by the user
- 20x2 Liquid Crystal Display for standard model **[Display can be replace by 20×4 or 40×2 on request at additional cost]**
- 104/105KeysIBM PC Compatible ASCII Keyboard
- 8251 USART interface through RS-232C port
- 48 programmable I/O lines through 2 nos. of 8255
- 16 bit programmable timer/counter using 8253
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- RS-232C interface for CRT using SID / SOD lines
- Real Time Clock using 6264 (OPTIONAL).
- Two modes of Commands: Key board Mode & Serial Mode
- All address, data and control lines are buffered & made available at connector as per STD Bus configuration
- Powerful Software Commands like relocate, string, fill, insert, delete, block move, memory compare etc
- Onboard Single Line Assembler/Disassembler
- Facility for Downloading/Uploading files from/to PC
- **Enclosed in an attractive ABS plastic cabinet with cover**
- **In-built Power Supply**
- User's Manual

TECHNICAL SPECIFICATIONS FOR VMC-8509ADU

- All Specifications are same as **VMC-8509ADU** with additional features as given below :
- Additional serial port using 8251
- EPROM Programmer for 27 series with 28 pin ZIF
- Interrupt Controller using 8259 & Printer Interface
- 8 Channel 8 bit ADC using 0809
- 1 Channel 8 bit D/A using DAC 0800
- 1 Relay output & 1 Opto isolated input

ORDERING GUIDE :

VMC-8509	WITH RS-232 SERIAL INTERFACE
VMC-8509U	WITH USB & RS-232C SERIAL INTERFACE
VMC-8509AD	ONBOARD ADC, DAC, RELAY, OPTO & EPROM PROGRAMMER
VMC-8509ADU	WITH USB INTERFACE & ONBOARD ADC, DAC, RELAY, OPTO & EPROM PROGRAMMER

ORDER 8506 instead of 8509 for Seven Segment Display & Hex keypad instead of LCD display & PC keyboard

DISTUDENT-85U DISTUDENT-85ADU

8085 MICROPROCESSOR TRAINER WITH LCD DISPLAY, USB INTERFACE & INBUILT POWER SUPPLY

TECHNICAL SPECIFICATIONS FOR STUDENT-85U

- Based on 8085 CPU operating at 6.144 MHz crystal frequency
- 8K bytes of EPROM loaded with powerful monitor program
- 8K bytes of RAM available to the user
- Total on board memory expansion upto 64K bytes
- Memory mapping definable by the user
- 16 bit programmable TIMER/COUNTER using 8253
- 48 programmable I/O lines provided through 8255
- RS-232C interface through SID/SOD Lines with Auto baud rate
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- Two modes of commands:
- - Hex Key pad mode & Serial mode
- All address, data & control lines are buffered and made available at the edge connector as per STD bus configuration
- IBM PC Compatible ASCII Keyboard
- 16x2 LCD display for standard model **[Display can be replace by 20×2 or 20×4 or 40×2 on request at additional cost]**
- Powerful software commands like Relocate, String, Fill, Insert, Delete, Block Move, Examine/Compare Memory, Examine Register, Insert Data, Delete Data, Single Step, GO, Break Point in both Serial and Keyboard mode
- Facility for Downloading/Uploading files from/to PC
- On board Disassembler
- Battery back up provided
- Enclosed in an attractive ABS plastic cabinet with cover
- In-Built Power Supply
- User's Manual

TECHNICAL SPECIFICATIONS FOR STUDENT-85ADU

- All Specifications are same as **STUDENT-85U** with additional features as given below :
- I/O lines Controlled
- 8 LED using 8 nos. of I/O lines
- 2 Relay of 12V & 1 Opto
- 8 Bit DAC using DAC-0800
- 8 Channel 8 Bit ADC using ADC-0809

ORDERING GUIDE :

STUDENT-85	WITH RS-232 SERIAL INTERFACE
STUDENT-85U	WITH USB & RS-232 SERIAL INTERFACE
STUDENT-85AD	ONBOARD ADC, DAC, RELAY, OPTO & I/O LINES USING LED
STUDENT-85ADU	WITH USB INTERFACE & ONBOARD ADC, DAC, RELAY, OPTO & I/O LINES USING LED

Specifications are subject to change without notice due to our constant efforts for improvement.

PRODUCT SELECTION GUIDE

DYNAMIC INDUSTRIES

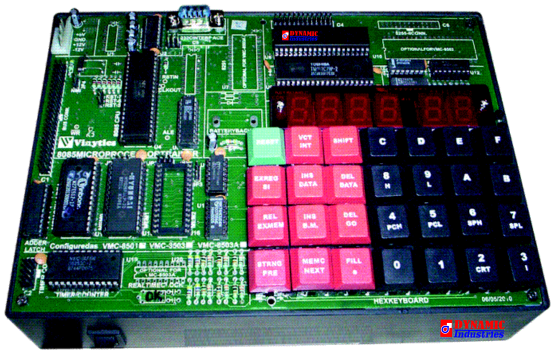
L-169A, Lajpat Nagar, Sahibabad, Ghaziabad, Uttar Pradesh – 201005
Email: dynamicindustries05@gmail.com,Phone no. 8218385397

Microprocessor/Microcontroller Lab

DYNAMIC Industries

DIVMC-8501U DIVMC-8609U DIVMC-8609ADU DIVMCVMC-8502U

8085 MICROPROCESSOR TRAINER WITH LED DISPLAY, USB INTERFACE & INBUILT POWER SUPPLY



TECHNICAL SPECIFICATIONS FOR VMC-8501U

- Based on 8085 CPU operating at 6.144 MHz crystal frequency
- 8K bytes of EPROM loaded with powerful monitor program
- 8K bytes of RAM available to the user
- Total on board memory expansion upto 64K bytes using 2732/2764/27128/6264/ 62256 with 1 socket of 28 pin
- Battery back up for RAM with 3.6V rechargeable cell
- Memory mapping definable by the user
- 16 bit programmable TIMER/COUNTER using 8253
- Additional serial port using 8251 **(OPTIONAL)**
- Real Time Clock **(OPTIONAL)**
- 24 I/O lines provided through 8255
- RS-232C interface through SID/SOD Lines with Auto baud rate
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- Two modes of commands:
  - Hex Key pad mode & Serial mode
- All address, data & control lines are buffered and made available at the edge connector as per STD bus configuration
- 25/28 key hexadecimal keyboard and six seven segment displays through 8279
- Powerful software commands like Relocate, String, Fill, Insert, Delete, Block Move, Examine/Compare Memory, Examine Register, Insert Data, Delete Data, Single Step, GO, Break Point in both Serial & Keyboard mode
- Facility for Downloading/Uploading files from/to PC
- **Enclosed in an attractive ABS plastic cabinet with cover**
- **In-built Power supply**
- User's Manual

TECHNICAL SPECIFICATIONS FOR VMC-8502U

- All Specifications are same as **VMC-8501U** with additional features as given below:
- 22 I/O lines provided through 8155

ORDERING GUIDE :

VMC-8501	WITH RS-232 SERIAL INTERFACE
VMC-8501U	WITH USB & RS-232 SERIAL INTERFACE
VMC-8502	ADDITIONAL I/O LINES THROUGH 8155
VMC-8502U	WITH USB INTERFACE & ADDITIONAL I/O LINES THROUGH 8155
VMC-8503	48 I/O LINES THROUGH 8255
VMC-8503U	WITH USB INTERFACE & 48 I/O LINES THROUGH 8255

8086/88 MICROPROCESSOR TRAINER WITH LCD DISPLAY, USB INTERFACE, EPROM PROGRAMMER, ADC, DAC & INBUILT POWER SUPPLY

TECHNICAL SPECIFICATIONS FOR VMC-8609U

- Based on 8086/8088 Microprocessor
- Onboard assembler & disassembler
- 16K Bytes of EPROM Loaded with monitor expandable to 256K Bytes using 27256 with commands like Assemble, Display or Modify Data, Unassembled, Trace, Go
- 16K bytes of CMOS RAM expandable to 128K Bytes using 6264/62256
- 72 I/O lines using three nos. of 8255
- 8 different level interrupt through 8259
- Three 16 bit Timer/Counter through 8253
- 104/105Keys IBM PC Compatible ASCII Keyboard
- 20x2 Liquid Crystal Display for standard model
  [Display can be replace by 20×4 or 40×2 on request at additional cost]
- RS-232C Port using 8251
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- All address, data and control signals (TTL Compatible) available at edge connector as per Multi Bus. The kit also has its own Resident Bus
- **Enclosed in an attractive ABS plastic cabinet with cover**
- **In-built Power Supply**
- User's Manual

TECHNICAL SPECIFICATIONS FOR VMC-8609ADU

- All Specifications are same as **VMC-8609U** with additional features as given below:
- On board 8 channel 8 Bit ADC
- On board Single channel 8 bit DAC
- Real time clock (optional)
- EPROM Programmer (Optional)

TECHNICAL SPECIFICATIONS FOR VMC-8609NIOU

- All Specifications are same as **VMC-8609ADU** with additional features as given below:
- Onboard sockets provided to facilitate the use of 8089 I/O processor & 8087 Numeric Data Processor with printer interface

ORDERING GUIDE :

VMC-8609	WITH RS-232 SERIAL INTERFACE
VMC-8609U	WITH USB &RS-232 SERIAL INTERFACE
VMC-8609AD	ONBOARD ADC, DAC, RELAY & OPTO
VMC-8609ADU	WITH USB INTERFACE & ONBOARD ADC, DAC, RELAY & OPTO
VMC-8609NIO	NUMERIC DATA & I/O PROCESSOR
VMC-8609NIOU	WITH USB INTERFACE, NUMERIC DATA & I/O PROCESSOR

PRODUCT SELECTION GUIDE

Specifications are subject to change without notice due to our constant efforts for improvement.

DYNAMIC INDUSTRIES

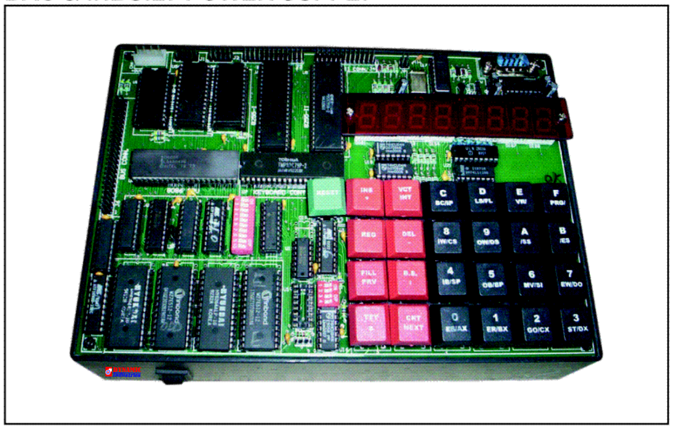
L-169A, Lajpat Nagar, Sahibabad, Ghaziabad, Uttar Pradesh – 201005
Email: dynamicindustries05@gmail.com,Phone no. 8218385397

Microprocessor/Microcontroller Lab

DYNAMIC Industries

DIVMC=8603U DIVMC=8603ADU

8086/88 MICROPROCESSOR TRAINER WITH LED DISPLAY, USB INTERFACE, EPROM PROGRAMMER, ADC, DAC & INBUILT POWER SUPPLY



TECHNICAL SPECIFICATIONS FOR VMC-8603U

- Based on Intel's 8086/8088 CPU operating at 2.5/5MHz
- 16K bytes of RAM available to the user using 6264
- 16K bytes of EPROM loaded with powerful monitor program (2764/27128)
- Total on board memory capacity of 128KB of RAM & 128KB of EPROM
- Battery backup provided for RAM area
- 48 I/O lines through 2 nos. of 8255
- 16 bit Timer/Counter through 8253
- RS-232C for CRT Terminal through 8251 Baud rate selection through DIP switch from 110-19,200 baud
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- Provision for EPROM Programmer for 2732/2764/27256 with 28 pin ZIF with facility to Program, Verify, List & Blank Check for Even, Odd & Continuous bytes
- 25/28 keys hexadecimal keyboard & eight seven segment display.
- Resident Monitor with two modes of operation:
  - Keyboard mode & Serial mode
- Powerful software commands like GO, EXAMINE/MODIFY REGISTERS, SINGLE STEPPING, BLOCK MOVE, FILL, INSERT, DELETE, INPUT/OUTPUT BYTE & WORD
- Facility for uploading/downloading of files from/to PC
- All Address, Data and Control Signals, are buffered and available at the FRC connector
- **Enclosed in an attractive ABS plastic cabinet with cover**
- **In-built Power Supply**
- User's Manual

TECHNICAL SPECIFICATIONS FOR VMC-8603ADU

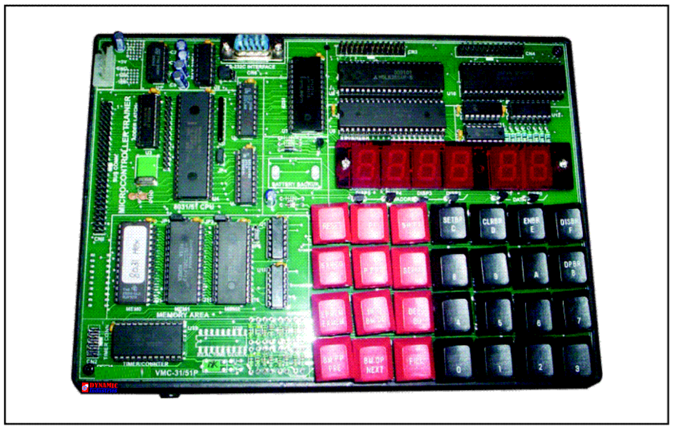
- All Specifications are same as **VMC-8603ADU** with additional features as given below :
- On board Analog to Digital Converter
- On board Digital to Analog Converter
- EPROM Programmer
- Onboard sockets provided to facilitate the use of 8089 I/O processor & 8087 Numeric Data Processor (Optional)

ORDERING GUIDE :

D VMC-8603U	WITH USB & RS-232 SERIAL INTERFACE
I VMC-8603	WITH RS-232 SERIAL INTERFACE
VMC-8603AD	ONBOARD ADC, DAC & EPROM PROGRAMMER
DIVMC-8603ADU	WITH USB INTERFACE & ONBOARD ADC, DAC & EPROM PROGRAMMER

DIVMC-31/51U DIVMC-31/51

8031/51 MICROCONTROLLER TRAINER KIT WITH LED DISPLAY, USB INTERFACE, & INBUILT POWER SUPPLY



TECHNICAL SPECIFICATIONS FOR VMC-31/51U

- Based on 8031/8051/8751 operating at 10/12 MHz.
- On board 8K RAM.
- Battery backup for RAM area.
- 8/16K bytes of EPROM with powerful monitor program.
- Total memory expandable upto 128K Bytes using four 28 pin sockets.
- 48 I/O lines using 2 nos. of 8255.
- Two External interrupts INT0 & INT1.
- Hexadecimal Keyboard & six Seven Segment displays.
- RS-232C interface using 8251.
- Auxiliary RS-232C using serial pins of 80C31.
- All data, address and control signals (TTL compatible) available at FRC connector.
- Powerful software commands like INSERT, DELETE, BLOCK MOVE, SET / CLEAR BREAK POINT, SINGLE STEP, EXAMINE THROUGH REGISTER, EXECUTE, EXAMINE, MODIFY, PROGRAM / DATA/INTERNAL MEMORY etc.
- Uploading/Downloading facility from PC in Intel Hex format.
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- **Enclosed in an attractive ABS plastic cabinet with cover**.
- **In-built Power Supply**
- User's Manual.

ORDERING GUIDE :

DIVMC-31/51	WITH RS-232 SERIAL INTERFACE
DIVMC-31/51U	WITH USB & RS-232 SERIAL INTERFACE

PRODUCT SELECTION GUIDE

Specifications are subject to change without notice due to our constant efforts for improvement.

DYNAMIC INDUSTRIES

L-169A, Lajpat Nagar, Sahibabad, Ghaziabad, Uttar Pradesh – 201005
Email: dynamicindustries05@gmail.com,Phone no. 8218385397

Microprocessor/Microcontroller Lab

DYNAMIC Industries

PRODUCT SELECTION GUIDE

DIVMC-31/51LCDU DIVMC-31/51LCDADU

8031/51 MICROCONTROLLER TRAINER KIT WITH LCD DISPLAY, USB INTERFACE, ADC, DAC & INBUILT POWER SUPPLY

TECHNICAL SPECIFICATIONS FOR VMC-31/51LCDU

- Based on 8051/89C52 Microcontroller
- On board 32K RAM
- Battery backup for RAM area
- 64 K bytes of EPROM with powerful monitor program
- Total memory expandable upto 256K Bytes using four 28 pin sockets
- 3 channel programmable timer/counter using 8253
- IBM PC compatible keyboard
- 20 x 2 LCD display for standard model
  [Display can be replace by 20×4 or 40×2 on request at additional cost]
- 48 I/O lines using 2 nos. of 8255
- IBM PC keyboard
- Two External interrupts INT0 & INT1
- RS-232C interface using 8251
- One serial USART interface provided by 89C51/89C52
- Auxiliary RS-232C using serial pins of 80C31
- **USB Interface to connect kit to computer (OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- All data, address and control signals (TTL compatible) available at FRC connector
- Powerful software commands like INSERT, DELETE, BLOCK MOVE, SET / CLEAR BREAK POINT, SINGLE STEP, EXAMINE THROUGH REGISTER, EXECUTE, EXAMINE, MODIFY, PROGRAM / DATA/INTERNAL MEMORY etc
- Uploading/Downloading facility from PC in Intel Hex format
- **Enclosed in an attractive ABS plastic cabinet with cover**
- **In-built Power Supply**
- User's Manual.

TECHNICAL SPECIFICATIONS FOR VMC-31/51LCDADU

- All Specifications are same as **VMC-31/51LCDU** with additional features as given below :
- Onboard Analog to Digital converter using ADC0809
- Onboard Digital to Analog converter using DAC0800
- Onboard Real Time Clock

ORDERING GUIDE :

DIVMC-31/51LCD	WITH RS-232 SERIAL INTERFACE
DIVMC-31/51LCDU	WITH USB & RS-232 SERIAL INTERFACE
	ONBOARD ADC, DAC
	WITH USB INTERFACE, ADC & DAC

ORDER VMC-89C52 instead of VMC-31/51LCD for 89C52 Microcontroller instead 8031/8051

DIVMC-89C61U

ADVANCED MICROCONTROLLER TRAINER KIT WITH 89C61/89C52 & USB INTERFACE

TECHNICAL SPECIFICATIONS

- 89C61/89C52 CPU operating @ 11.0592 MHz crystal
- 32K user RAM using 62256 with Battery Backup using NICD Battery
- 16K bytes of powerful monitor EPROM using 27512
- One memory socket is provided for expansion up to 64k
- On-board ISP programming facility with windows based ISP software
- 48 I/O lines using 8255 brought at 26 Pins FRC Connector
- Three Channel Timer/Counter using 8253
- On-Board 12Bit ADC using AD574
- On-Board 1Ch. DAC using DAC0800
- On-Board Real Time Clock using RTC6242 (OPTIONAL)
- AUX. RS-232C Interface using 8251 terminated on 9 Pins D-Type Connector
- 20x2 Alphanumeric LCD Display with Backlite for standard model
  [[Display can be replace by 20×4 or 40×2 on request at additional cost]
- **USB Interface to connect kit to computer(OPTIONAL)**
- **V-USB Terminal Software** to operate kit from USB Port to execute all commands like Examine Mem, Execute, Assembler & Dissembler etc
- 101 ASCII Keyboard interface using 89C2051 operating @ 12MHz
- Two External interrupts INT0 & INT1 are available at FRC connector
- RS-232C using RX/TX of 8051 terminated on 9 Pins D-Type Connector
- Onboard Single Line Assembler / Disassembler
- Two modes of operation:
  ASCII Keyboard Mode & Serial Mode
- Powerful Commands like Examine/Edit Memory, Examine/Edit Register, Single stepping, Execution, Break Point can be used through ASCII keyboard or PC serial mode
- Facility for Downloading/Uploading files from/to PC
- All Address, Data, Control & Port lines are available on 50 Pins FRC Connector
- **In-Built Power Supply**
- **Enclosed in an attractive ABS plastic cabinet with cover**
- User's Manual with sample program

ORDERING GUIDE :

DIVMC-89C61	WITH RS-232 SERIAL INTERFACE
DIVMC-89C61U	WITH USB & RS-232 SERIAL INTERFACE

Specifications are subject to change without notice due to our constant efforts for improvement.

DYNAMIC INDUSTRIES

L-169A, Lajpat Nagar, Sahibabad, Ghaziabad, Uttar Pradesh – 201005
Email: dynamicindustries05@gmail.com,Phone no. 8218385397